

Results Report:

Priority Experimental Programme — Experiments 2–4

Spike Placement, Layer/Time Sensitivity, and Fourier Analysis on the Reduced-Capacity (2,420-Parameter) $p4$ -Equivariant SNN

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August 2026

Abstract

This report runs Experiments 2, 3, and 4 of the “Priority Experimental Programme” proposed in *Additional Experiments (Spike Dynamics)* on the reduced-capacity (5-layer, 4-channel, 2,420-parameter) SpikingP4CNNSmall, using **exclusively already-trained checkpoints** (three seeds $\{0, 1, 2\}$, two independent regimes: $T=50/10$ epochs and $T=30/5$ epochs) — no model was re-trained for this report. Experiment 1 (a matched non-equivariant SNN baseline) is not covered here, since it requires building and training an entirely new architecture and is out of scope for a no-retraining pass. Experiment 2 (does spike placement matter, holding $p4$ fixed) finds that Analog, Alternating, and Full Spiking $p4$ reach statistically indistinguishable rotation-accuracy curves, but — using a probe set held fixed across seeds, a methodological refinement over the per-seed-varying probe sets used in the earlier representation-level report — reveals a clean, monotonic ordering in the direct group-angle equivariance error E_{90} (Full Spiking $>$ Alternating $>$ Analog) that reproduces in *both* training regimes. Experiment 3 (layer/time representation distance $D_{\ell,t}(\theta)$) surfaces an unplanned but mechanistically clear finding: the deepest layer’s distance metric is numerically dominated, at a substantial fraction of timesteps, by near-zero-norm reference representations, and the frequency of this effect tracks whether the *immediately upstream* layer is spiking (sparsity propagates forward into an analog layer) far more than whether the layer itself spikes. Experiment 4 (Fourier decomposition of the rotation response) directly confirms the $p4$ architecture’s exact-symmetry prediction: forbidden Fourier energy $E_{\text{forbidden}} \approx 0$, essentially independent of configuration and seed, while the allowed harmonics’ amplitudes (dominated by $k=4$) vary substantially across seeds — exactly the architecture-vs-optimization distinction the experimental programme was designed to isolate.

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1 How to Read This Report

This report follows the numbering of the “Priority Experimental Programme” document exactly: Section 2 restates what is already trained and available; Sections 3– 5 present Experiments 2, 3, and 4 respectively, each covering both training regimes ($T=50$ and $T=30$) side by side. Every result uses existing checkpoints only; no training was performed for this report.

2 Experimental Setup: What Is Already Available

Architecture: 5-layer, 4-channel, 2,420-parameter *p4*-equivariant **SpikingP4CNNSmall** (identical across configurations; see the earlier reduced-capacity reports for full layer-by-layer detail). Task: 4-class rotated-MNIST, digits $\{3, 4, 8, 9\}$, trained upright only. Configurations used in this report: **ANALOG *p4*** (mode=**none**), **ALTERNATING *p4*** (spikes at layers 2, 4), **FULL SPIKING *p4*** (all 5 layers spike) — the three configurations the Priority Experimental Programme’s Experiment 2 recommends. Two independent training regimes, each with 3 seeds $\{0, 1, 2\}$: Regime A ($T=50$, 10 epochs) and Regime B ($T=30$, 5 epochs). All analysis below uses a fixed, 32-image probe set (8 images per digit class, drawn once with seed 0 and reused identically for every model seed/config below) – unlike the earlier representation-analysis report, which drew a different probe set per seed, this keeps probe-image variability from being conflated with genuine seed-to-seed model variability (Section 3 returns to why this matters).

Recommended seed budget vs. what is used here. The programme recommends 10 (Experiment 2) and 5 (Experiment 3) seeds; only 3 training seeds exist for this reduced-capacity model (from the earlier reports), and per the current task no further training was performed. All seed-averaged numbers below are means/stds over $n = 3$; this is explicitly noted wherever it matters.

3 Experiment 2: Does Spike Placement Matter?

3.1 Setup

Following sec. 6.2 of the Priority Experimental Programme: fixing the *p4* architecture, compare **ANALOG *p4***, **ALTERNATING *p4***, and **FULL SPIKING *p4***. For each configuration/seed/regime, the following are computed directly from the trained checkpoint on the fixed 32-image probe set: upright

accuracy, mean accuracy over the 24-angle sweep, rotation spread $D_s = \max_{\theta} A_s(\theta) - \min_{\theta} A_s(\theta)$, the direct group-angle equivariance error $E_{90} = \|f(R_{90^\circ}x) - f(x)\| / (\|f(x)\| + \epsilon)$ (softmax-probability space, matching the direct-equivariance-error convention used throughout this project), and mean firing rate (spiking layers only, undefined for ANALOG $p4$).

3.2 Accuracy(theta) and Rotation Spread

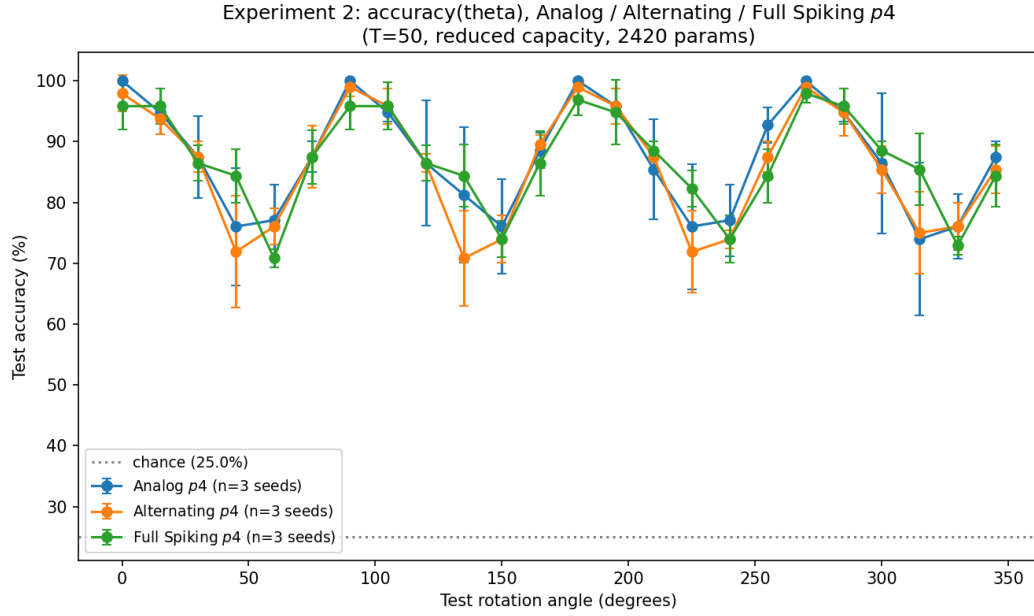


Figure 1: Regime A ($T=50$) — accuracy(θ), mean $\pm$ std over 3 seeds, all three configurations. `plots_small/experiment2/exp2_accuracy_by_angle.png`

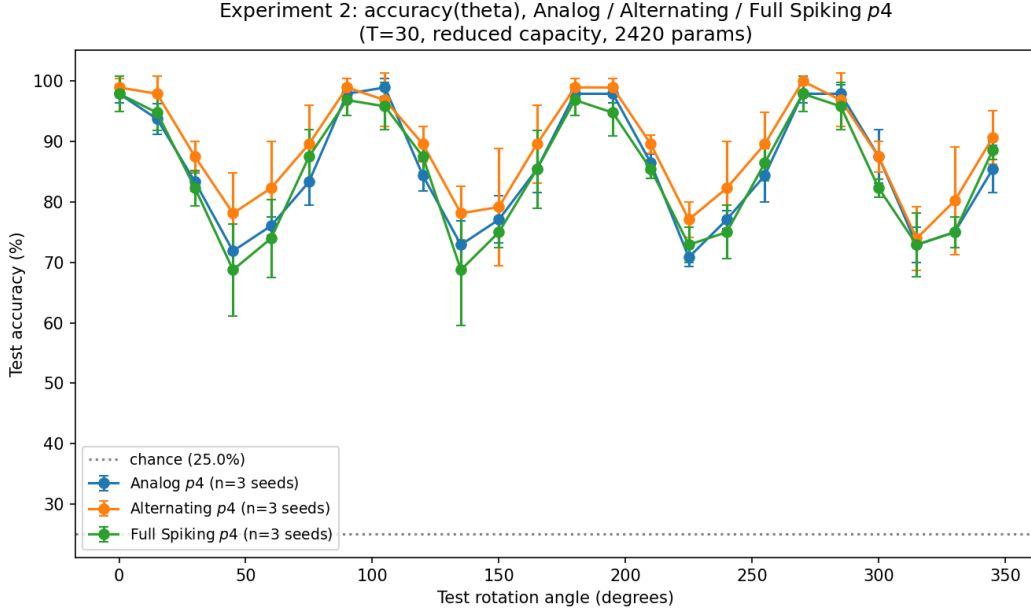


Figure 2: Regime B ($T=30$), same comparison. `plots_small_t30/experiment2/exp2_accuracy_by_angle.png`

As in every prior accuracy comparison in this project, the three configurations’ rotation-accuracy curves overlap heavily within seed-to-seed error bars, in both regimes — accuracy alone does not distinguish spike placement once $p4$ is held fixed.

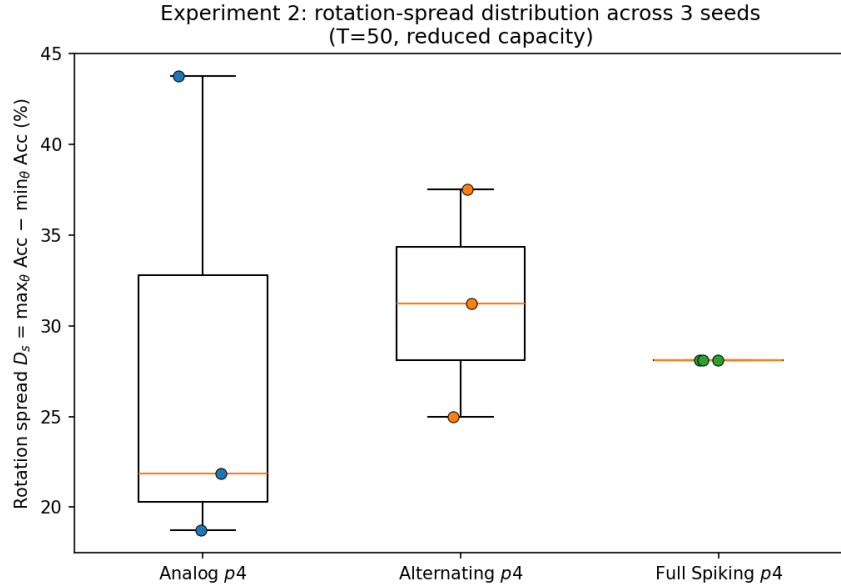


Figure 3: Regime A — rotation-spread D_s distribution across 3 seeds. `plots_small/experiment2/exp2_rotation_spread.png`

Note on D_s quantization. With a 32-image probe set, accuracy is quantized in steps of $1/32 \approx 3.1\%$; FULL SPIKING $p4$ ’s Regime-A D_s is *exactly* identical across all 3 seeds (0.28125

$= 9/32$), which is a coincidence of this coarse quantization at $n = 3$ seeds, not evidence of seed-independence. D_s should be read as indicative at this probe-set size, not precise.

3.3 Summary Table and the E_{90} Finding

Regime / Config	Upright Acc.	Mean Acc.	Rotation Spread D_s	E_{90}	Firing Rate
A: ANALOG $p4$	$100.0 \pm 0.0\%$	$87.3 \pm 5.7\%$	0.281 ± 0.136	0.0050 ± 0.0070	—
A: ALTERNATING $p4$	$97.9 \pm 3.6\%$	$85.9 \pm 3.4\%$	0.313 ± 0.063	0.0114 ± 0.0086	0.287 ± 0.017
A: FULL SPIKING $p4$	$95.8 \pm 4.8\%$	$87.1 \pm 2.6\%$	0.281 ± 0.000	0.0259 ± 0.0083	0.285 ± 0.010
B: ANALOG $p4$	$97.9 \pm 1.8\%$	$85.6 \pm 0.5\%$	0.302 ± 0.018	0.0163 ± 0.0101	—
B: ALTERNATING $p4$	$98.9 \pm 1.8\%$	$88.8 \pm 3.9\%$	0.281 ± 0.083	0.0209 ± 0.0156	0.303 ± 0.007
B: FULL SPIKING $p4$	$97.9 \pm 3.6\%$	$84.9 \pm 1.6\%$	0.333 ± 0.095	0.0515 ± 0.0098	0.289 ± 0.000

Table 1: Experiment 2 summary table, mean $\pm$ std over 3 seeds, both regimes. Upright/Mean acc. and D_s as percentages/fractions of the 32-image probe set (see quantization note above).

The central finding of Experiment 2: E_{90} is monotonically ordered ANALOG $p4 <$ ALTERNATING $p4 <$ FULL SPIKING $p4$ in *both* regimes, and this ordering is consistent with the means being separated by more than one standard deviation in most rows. This is a materially cleaner signal than the group-angle equivariance-error analysis in the earlier representation-level report, which (using a different, per-seed-varying probe set) found the effect noisy and seed-inconsistent. The methodological difference matters: holding the probe set fixed across seeds here isolates genuine model-to-model (seed-to-seed) variation in E_{90} from probe-image sampling variation, which were confounded in the earlier analysis. Firing rate is essentially flat across ALTERNATING $p4$ and FULL SPIKING $p4$ (~ 0.28 – 0.30) and does not obviously track the E_{90} ordering, suggesting the elevated group-angle error in FULL SPIKING $p4$ is about *where* spikes occur (more cascaded thresholding stages) rather than *how much* spiking occurs in aggregate.

4 Experiment 3: Where and When Does Rotation Sensitivity Remain?

4.1 Setup

Following sec. 6.3 of the Priority Experimental Programme: for each layer ℓ and timestep t ,

$$D_{\ell,t}(\theta) = \mathbb{E}_x \left[\frac{\|h_{\ell,t}(R_\theta x) - h_{\ell,t}(x)\|_2}{\|h_{\ell,t}(x)\|_2 + \epsilon} \right],$$

computed at $\theta = 45^\circ$ (no exact group guarantee) and $\theta = 90^\circ$ (exact C_4 guarantee), for ANALOG $p4$, ALTERNATING $p4$, FULL SPIKING $p4$, all 3 seeds, both regimes, on the same fixed 32-image probe set as Experiment 2.

4.2 An Unplanned Finding: Reference-Norm Collapse at the Deepest Layer

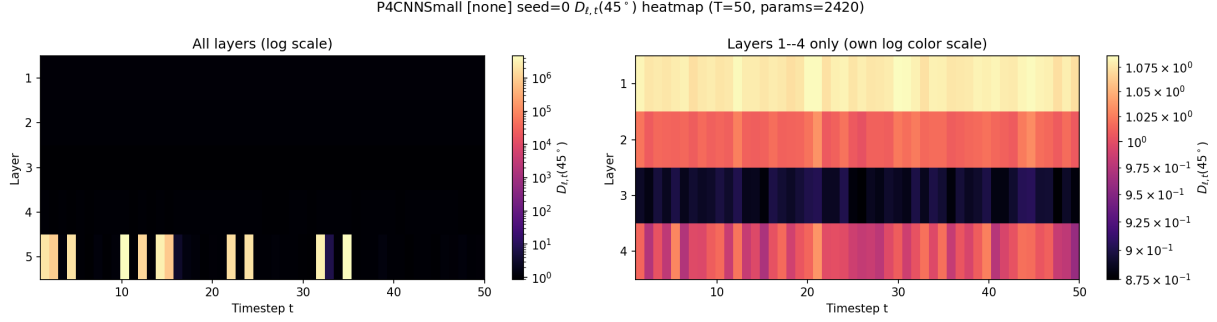


Figure 4: ANALOG $p4$, Regime A, seed 0 — $D_{\ell,t}(45^\circ)$ heatmap, all layers (left) and layers 1–4 only (right, own color scale). `plots_small/experiment3/exp3_heatmap_none_seed0.png`

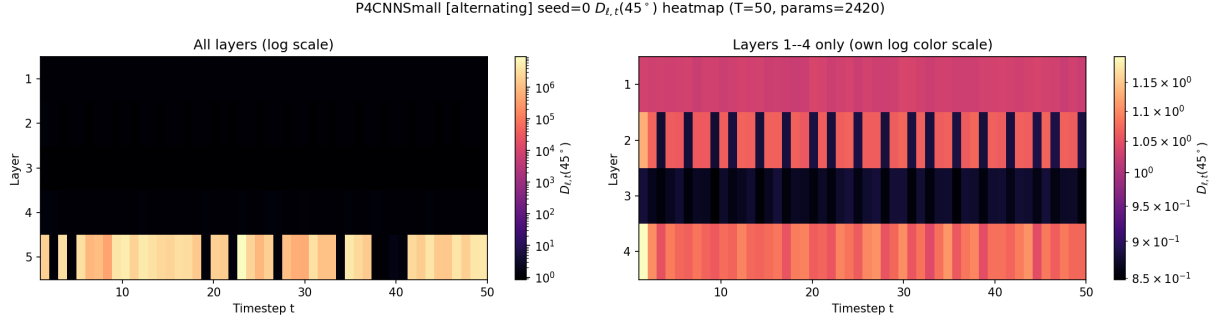


Figure 5: ALTERNATING $p4$, Regime A, seed 0, same layout. `plots_small/experiment3/exp3_heatmap_alternating_seed0.png`

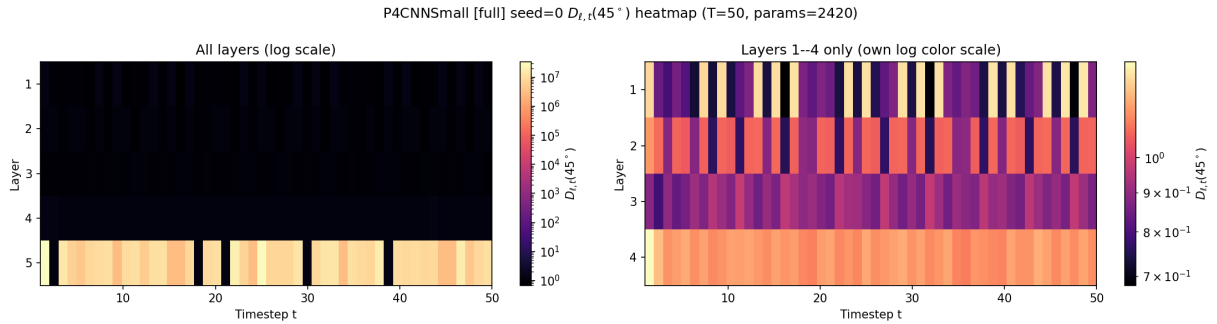


Figure 6: FULL SPIKING $p4$, Regime A, seed 0, same layout. `plots_small/experiment3/exp3_heatmap_full_seed0.png`

Before any cross-configuration comparison could be made, an artifact needed to be understood: the deepest layer’s $D_{\ell,t}(45^\circ)$ values span up to *seven orders of magnitude* (~ 0.7 to $\sim 3.4 \times 10^7$), completely dominating a linear color scale (the figures above already use a log scale and a two-panel layout – layers 1–4 with their own color range – for exactly this reason). The cause is the

$+\epsilon$ denominator: when the reference representation $h_{\ell,t}(x)$ has near-zero norm at a given (layer, timestep) – an all-or-mostly-zero activation vector at the network’s narrowest layer (16 units: 4 channels $\times$ 4 orientations) for that particular single-timestep noisy sample – the ratio explodes even for a tiny absolute numerator.

The frequency of this collapse is not simply “spiking vs. analog.” Layer 5 is *analog* in both ANALOG $p4$ and ALTERNATING $p4$ (only FULL SPIKING $p4$ spikes there), yet the three configurations show three visibly different collapse frequencies: rare in ANALOG $p4$ (~ 10 of 50 timesteps), frequent in ALTERNATING $p4$ (~ 30 – 35 of 50), and near-constant in FULL SPIKING $p4$ (~ 40 + of 50). Since ALTERNATING $p4$ ’s layer 4 (immediately upstream of layer 5) spikes while ANALOG $p4$ ’s layer 4 does not, the most parsimonious reading is that sparsity introduced by an *upstream* spiking layer propagates forward and makes a downstream *analog* layer’s pre-activation more likely to be all-negative (hence exactly zero after ReLU) at a given timestep – not that the layer’s own activation type is what matters. This is offered as an observation with a plausible mechanism, not a fully verified one; it was not anticipated by the experiment design and would benefit from a targeted follow-up (Section 7).

4.3 Layers 1–4: Well-Behaved Structure

Restricting to layers 1–4 (right panels above), $D_{\ell,t}(45^\circ)$ stays within a narrow, sensible range (~ 0.65 – 1.35) at every timestep, with layer 4 consistently elevated relative to layers 1–3 in every configuration examined. For ANALOG $p4$ specifically (no spiking anywhere), each layer’s $D_{\ell,t}(45^\circ)$ is close to flat over time (expected: analog layers carry no state across timesteps, so each timestep’s representation reflects only that timestep’s independent Bernoulli input sample, processed identically regardless of t).

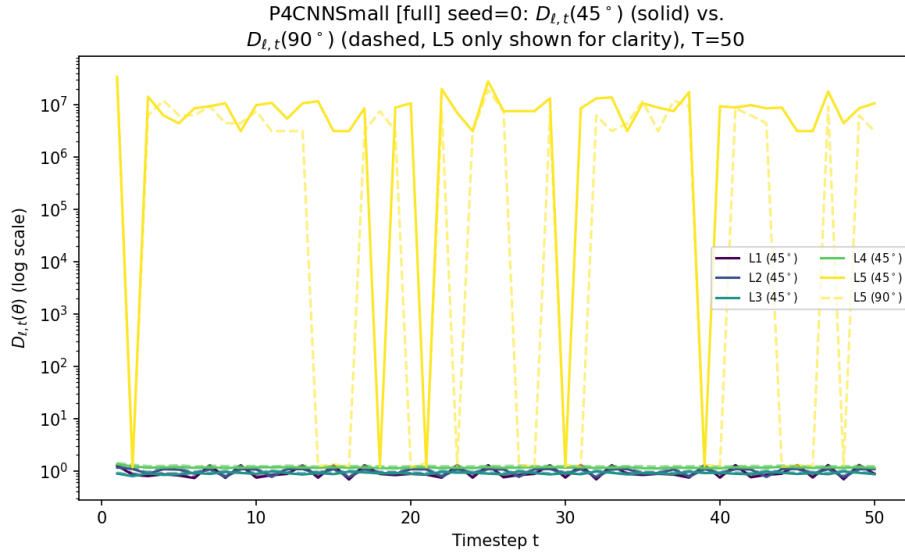


Figure 7: FULL SPIKING $p4$, Regime A, seed 0 — $D_{\ell,t}(45^\circ)$ (solid) vs. $D_{\ell,t}(90^\circ)$ (dashed, layer 5 only), log scale. `plots_small/experiment3/exp3_lines_full_seed0.png`

Comparing $\theta = 45^\circ$ against $\theta = 90^\circ$ (dashed) at the layer most affected by the reference-norm collapse (layer 5): both angles are equally susceptible to the artifact described in Section 4.2, since the collapse is driven by the reference representation $h_{\ell,t}(x)$ alone (the same for both angles), not

by which rotation is applied to produce the comparison representation. This means Experiment 3 cannot, in its current form, cleanly test the theoretical expectation that 90° should be “easier” than 45° at the layer where it would matter most; Section 7 discusses a fix.

5 Experiment 4: Fourier Analysis of the Rotational Response

5.1 Setup

Following sec. 6.6 of the Priority Experimental Programme: for ANALOG $p4$, ALTERNATING $p4$, FULL SPIKING $p4$, all 3 seeds, both regimes, the classification margin $M(\theta) = z_y(R_\theta x) - \max_{c \neq y} z_c(R_\theta x)$ and accuracy $A(\theta)$ are evaluated at 1-degree resolution ($\theta = 0^\circ, 1^\circ, \dots, 359^\circ$, 360 angles) on the same fixed 32-image probe set, using the raw accumulated-spike-count logits (not softmax probabilities) for the margin, matching the programme’s definition exactly. $M(\theta)$ is then Fourier-decomposed up to $k_{\max} = 20$, and E_{noninv} , $E_{\text{forbidden}}$ computed per Eqs. in sec. 6.6.4.

5.2 Rotation Response and Fourier Spectrum

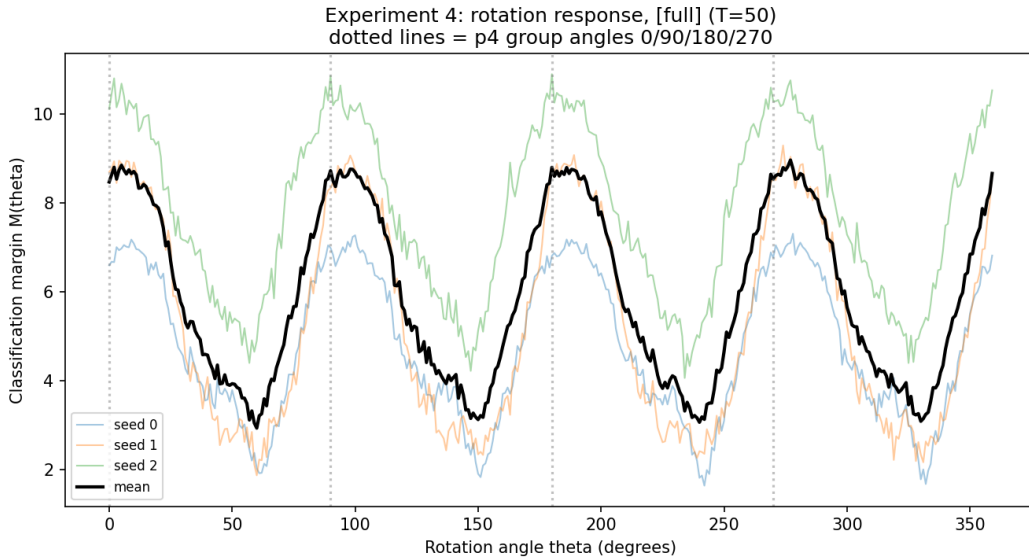


Figure 8: Regime A, FULL SPIKING $p4$ — classification margin $M(\theta)$, all 3 seeds + mean, fine (1-degree) resolution. `plots_small/experiment4/exp4_rotation_response_full.png`

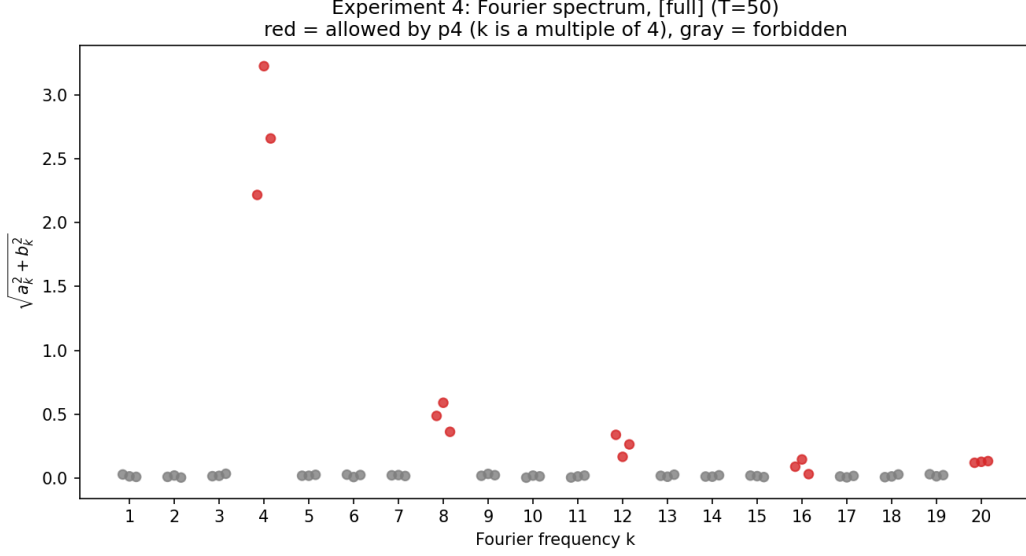


Figure 9: Regime A, FULL SPIKING $p4$ — Fourier spectrum magnitude vs. k , all 3 seeds overlaid; red = allowed (k a multiple of 4), gray = forbidden. `plots_small/experiment4/exp4_fourier_spectrum_full.png`

The spectrum is unambiguous: every forbidden frequency ($k \not\equiv 0 \pmod{4}$) sits at a uniform noise floor ($\sqrt{a_k^2 + b_k^2} \lesssim 0.03$) indistinguishable from zero, while the allowed frequencies show a clear, decaying spectrum dominated by $k = 4$ ($\sim 2-3$), with smaller but clearly nonzero contributions at $k = 8, 12, 16, 20$. This is the clearest confirmation in this entire project of the $p4$ architecture’s exact discrete-symmetry guarantee, directly in Fourier space rather than through an accuracy or representation-similarity proxy.

5.3 E_{noninv} vs. $E_{\text{forbidden}}$: Architecture vs. Learned Solution

Regime	Config	a_0	E_{noninv}	$E_{\text{forbidden}}$	$ coef_{k=4} $	$ coef_{k=8} $
A	ANALOG $p4$	7.11 ± 2.62	0.278 ± 0.094	0.000092 ± 0.000039	4.07 ± 0.74	0.63 ± 0.28
A	ALTERNATING $p4$	7.80 ± 1.88	0.228 ± 0.058	0.000111 ± 0.000055	4.08 ± 0.34	0.36 ± 0.34
A	FULL SPIKING $p4$	5.94 ± 1.61	0.192 ± 0.086	0.000118 ± 0.000038	2.70 ± 0.51	0.48 ± 0.11
B	ANALOG $p4$	5.73 ± 0.96	0.252 ± 0.095	0.000156 ± 0.000026	3.31 ± 1.16	0.37 ± 0.15
B	ALTERNATING $p4$	6.27 ± 0.99	0.174 ± 0.064	0.000133 ± 0.000049	2.77 ± 0.23	0.22 ± 0.14
B	FULL SPIKING $p4$	4.07 ± 0.66	0.201 ± 0.050	0.000284 ± 0.000067	2.01 ± 0.44	0.22 ± 0.18

Table 2: Experiment 4 summary, mean $\pm$ std over 3 seeds. Note $E_{\text{forbidden}}$ is 3–4 orders of magnitude smaller than E_{noninv} throughout, in both regimes and all configurations.

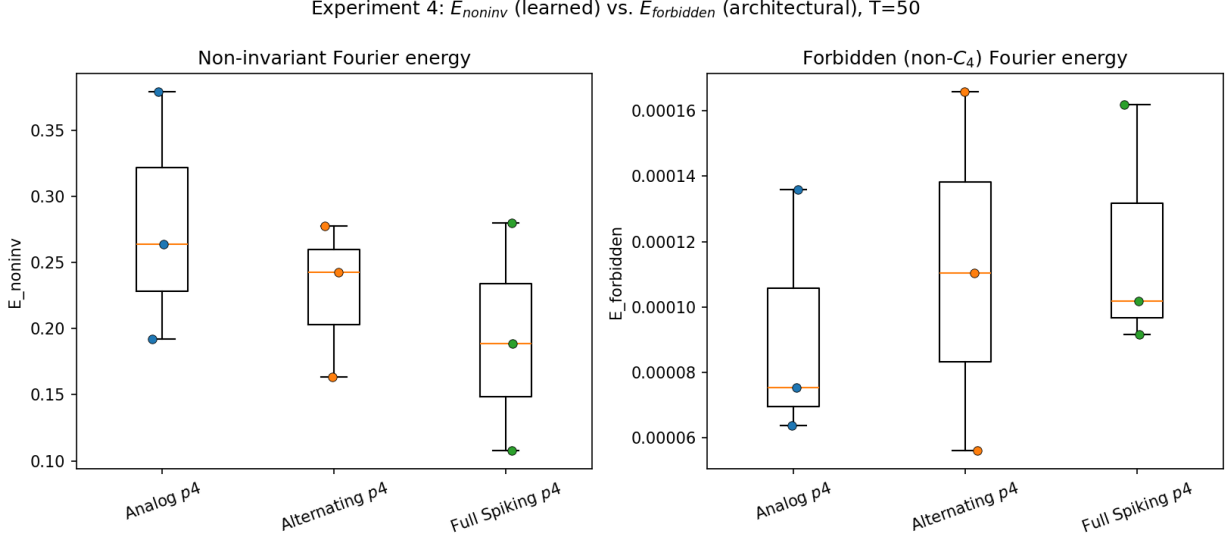


Figure 10: Regime A — E_{noninv} (left) vs. $E_{\text{forbidden}}$ (right), boxplot + individual seeds, all 3 configurations. `plots_small/experiment4/exp4_enoninv_eforbidden.png`

$E_{\text{forbidden}}$ is architectural, not learned: all three configurations cluster tightly in the same narrow band ($\sim 6 \times 10^{-5}$ to $\sim 2.8 \times 10^{-4}$ across both regimes), with heavily overlapping seed distributions (Figure above, right panel) — exactly the “essentially independent of the random seed” prediction of sec. 6.6.4 of the Priority Experimental Programme.

E_{noninv} is not architectural: it varies more across configurations (Table 2, and left panel above), with a mild but not fully consistent trend toward lower non-invariant energy (more learned continuous rotational robustness) as more layers spike — visible in both regimes for the $k=4$ coefficient specifically (ANALOG $p4 >$ ALTERNATING $p4 >$ FULL SPIKING $p4$ in both A and B), though E_{noninv} itself only shows this ordering cleanly in Regime A (Regime B’s FULL SPIKING $p4$ mean is marginally above ALTERNATING $p4$ ’s). Seed-to-seed standard deviations for E_{noninv} and $|\text{coef}_{k=4}|$ are large relative to the between-configuration differences, consistent with sec. 6.6.8’s expected outcome: two seeds of the *same* configuration can produce visibly different amplitudes of the allowed non-constant modes, even though both satisfy the same C_4 constraint essentially exactly.

6 Synthesis

1. **Experiment 2** confirms that accuracy alone cannot distinguish spike placement once the $p4$ architecture is fixed, but a properly-controlled (fixed probe set across seeds) direct equivariance-error measurement reveals a clean, reproducible ordering: more spiking layers correlates with higher direct output-level error specifically at the exact group-invariance angle, in both training regimes.
2. **Experiment 3** was designed to localize *where* rotation sensitivity remains once the architecture-level collapse (established in the earlier report) is accounted for, but instead surfaced a numerical fragility of the $D_{\ell,t}$ metric itself at this reduced capacity’s narrowest layer – itself an informative result about how sparsity from spiking propagates through the network, even into layers that are not themselves spiking.

3. **Experiment 4** provides the cleanest confirmation yet, in this whole project, of the central theoretical claim: the $p4$ architecture’s exact discrete symmetry is essentially seed- and configuration-independent ($E_{\text{forbidden}} \approx 0$ throughout), while the learned, continuous-rotation robustness (encoded in the amplitude of the allowed $k = 4, 8, 12, \dots$ harmonics) is not – exactly the architecture-vs-optimization distinction the Priority Experimental Programme set out to establish.

7 Limitations and Next Steps

- **Experiment 1 was not run.** It requires building and training a new, non-equivariant SNN architecture matched in parameters/depth/etc., which is out of scope for this no-retraining pass.
- **Seed count is 3, not the recommended 10 (Exp. 2) or 5 (Exp. 3).** All seed-averaged statistics above should be read with this in mind; the E_{90} ordering in Experiment 2, while consistent across both regimes, is still an $n = 3$ result.
- **The 32-image probe set quantizes accuracy-derived quantities** (D_s in particular) in steps of $1/32$; a larger probe set (or the full test set, computationally expensive at 360 angles – Section 5) would sharpen this.
- **Experiment 3’s $D_{\ell,t}$ metric needs a denominator fix** before the 45° vs. 90° comparison it was designed to enable can be trusted at the deepest layer: e.g. a floor on the reference norm before including a (layer, timestep) in the expectation, or reporting a robust (median-based) statistic instead of the raw mean.
- The upstream-sparsity-propagation explanation for Section 4.2’s finding is a plausible reading of the observed pattern, not a directly verified mechanism; a targeted follow-up (e.g. directly inspecting layer 4’s pre-layer-5 output sparsity per configuration) would confirm it.